Tree solutions

1

A tree is connected and acyclic. Deleting a leaf inductively from the one-vertex base proves $e=n-1$; equivalently connected graphs need at least $n-1$ edges and forests have at most that many. Conversely an acyclic graph with $e=n-1$ has $c=n-e=1$ by the forest formula, hence is connected and a tree.

2

$L=I+1$ gives $I=30$ and $e=I+L-1=60$. Since $2^4=16<31\le32=2^5$, height is at least five. Take a perfect height-five tree with 32 leaves and contract a deepest sibling-leaf pair to its parent leaf. It remains full, has 31 leaves, and still has height five, proving attainment.

3

The decision tree needs $6!=720$ leaves. Since $2^9=512<720\le1024=2^{10}$, its height is at least ten and $\lceil\log_2 720\rceil=10$.

Split into [5,1,4] and [2,6,3]. On the left split [5] and [1,4]; merging [1],[4] costs one, then compare 5 against 1 and 4 for two more, producing [1,4,5] in three. On the right, [6],[3] costs one; [2] versus [3,6] compares only 2,3 once, giving [2,3,6] in two total. Final comparisons are (1,2),(4,2),(4,3),(4,6),(5,6), then append six. Total is 3+2+5=10.

This is one input's cost. The split has worst-case $M(6)=2M(3)+5=2\cdot3+5=11$. Input [4,2,6,3,1,5] makes both halves use three and final merge five, attaining eleven. Matching the information lower bound once does not prove the algorithm optimal in worst case.

4

Merge 2+3=5, leaving 5(new),5,7,11; then 5+5=10, leaving 7,10,11; then 7+10=17; finally 17+11=28. Merge total is 5+10+17+28=60.

One code is 11:0,7:10,5:110,2:1110,3:1111. Its cost is $11\cdot1+7\cdot2+5\cdot3+2\cdot4+3\cdot4=60$. Mean is 60/28=15/7. Swapping left/right labels preserves these optimal lengths.

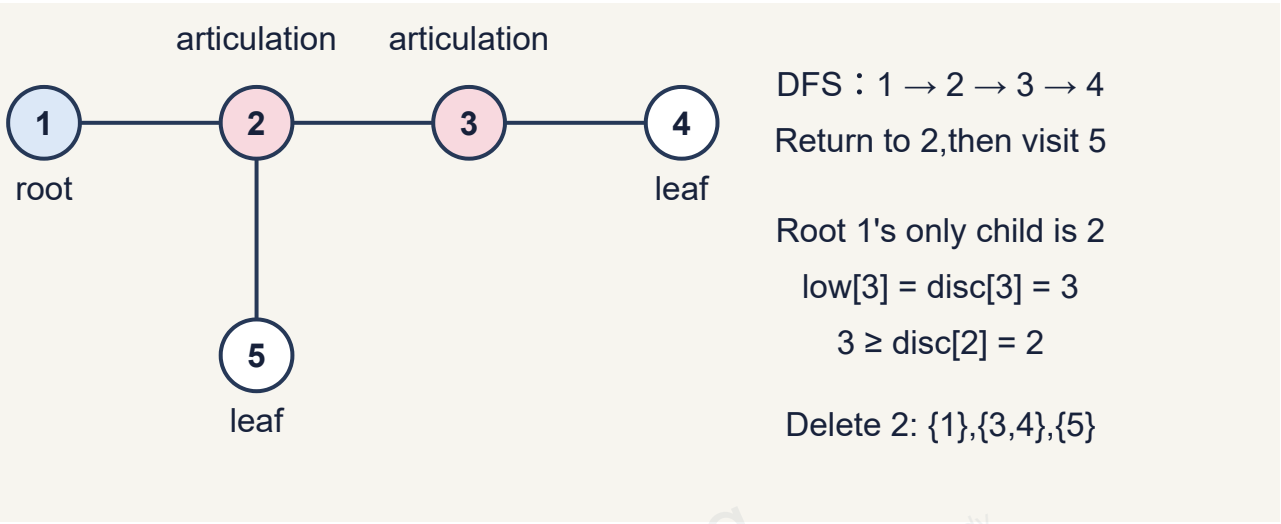
5

Connectivity supplies at least one path. If two distinct simple paths exist, take their first divergence and first later rejoining. The two internally disjoint intermediate paths form a simple cycle, contradicting acyclicity. A shortest nonzero closed walk alone is not enough: traversing one undirected edge out and back gives a length-two closed walk without a simple cycle. Thus there is exactly one simple path.

Tree solutions (continued)

6

Take 1–2–3–4 and add 2–5. With root one, its only child is two. Deleting nonroot two separates vertex one from the remaining branches; child three has low[3]≥disc[2] because no back edge reaches the root. Deleting three also separates four, so cuts are {2,3}. The root's one child does not prevent cuts elsewhere.



7

IDs are e_1=AB, e_2=BC, e_3=CA, e_4=CD, e_5=DE, e_6=EF, e_7=FD, e_8=DG. From A, the discovery order A, B, C, D, E, F, G has disc 1,2,3,4,5,6,7 and parents None, A, B, C, D, E, D.

Push 1,2,3,4,5,6,7. Back edge three makes low[C]=1; back edge seven makes low[F]=4,propagating low[E]=4. Returning E→D satisfies 4≥4: pop 7,6,5 for triangle DEF. Push eight; returning G→D satisfies 7≥4: pop eight for DG. Returning D→C satisfies 4≥3: pop four for CD. Returning C→B has 1<2:no pop. Returning B→A has 1≥1:pop 3,2,1 for ABC.

Final lows are 1,1,1,4,4,4,7. Nonroot C has child D with 4≥3; D has children E, G with lows≥4,so cuts are {C, D}. Strict inequality gives only bridges e_4,e_8. Every edge belongs to one of the four blocks, with triangles intact.

Rooting at D first explores C then A, B; after that E then F; finally G. Root D has three children C, E, G and deletion leaves ABC, EF, G, so it is a cut. C's AB subtree still cannot bypass C to D, so C remains a cut. Discovery times, lows, and pop order may change; cut set, bridges, and four blocks do not.

Tree solutions (continued)

8

Lengths (1,2,2,3) have Kraft sum 1/2+1/4+1/4+1/8=9/8>1,proving impossibility for every prefix code, rather than only failed guesses. Lengths (2,2,3,3,3,3) sum to 2/4+4/8=1 and permit `00,01,100,101,110,111`. First two start with zero and last four with one; within each group words are distinct and equal-length, hence prefix-free.

For one symbol with a separately known count,ε gives mean length zero. If every word must be nonempty, length at least one is necessary, and `0` attains it. Different specifications have different optima.

9

For m=3,l=13, n=ml+1=40,L=(m−1)l+1=27,e=39. The bound $27 \le 3^h$ gives $h \ge 3$. A perfect height-three ternary tree has levels 1,3,9,27,so internal count 1+3+9=13,leaves 27,and edges 3+9+27=39. This explicit tree attains height three.

10

Positive weights make unary internal nodes strictly wasteful: contraction decreases descendant depths. An optimal three-leaf tree is full, necessarily with depths (1,2,2). If weight w gets depth one, the cost is 2(x+y+z)−w. Choose largest z to minimize, giving W=2x+2y+z. Huffman merges x+y then x+y+z, giving the same sum. Words may be z:`0`,x:`10`,y:`11`.

Exchanging depth-two z with depth-one w changes cost by $w - z \le 0$,so keeping z deep cannot improve it. Ties allow several optima. For x=2,y=5,z=9,the merge weights are seven and sixteen, total 23,and mean 23/16=1.4375 bits.

11

If deleting tree edge uv left a u-to-v path, that alternative simple path plus uv would form a cycle. Thus every tree edge is a bridge. Handshake and e=n−1 give $\sum_v (\deg(v) - 2) = 2e - 2n = -2$.

At $n \ge 2$ no degree-zero vertex exists. Leaves contribute −L, degree-two vertices zero, three degree-three vertices contribute three, and one degree-five vertex three. Hence −L+3+3=−2,giving L=8.

Tree solutions (continued)

At n=1 the only vertex has degree zero and contributes −2,so the sum identity still holds. It is not a degree-one leaf; replacing all negative contributions by −L would miss this term. Under rooted no-children terminology the singleton is one leaf, a different convention.

12

Set disc[u]=1 and discover v on e_1 with disc[v]=2. Skip only parent-edge ID e_1. Edge e_2 still reaches ancestor u, so low[v]=min(2,1)=1. Upon return low[v]=disc[u],not strictly greater, and e_1 is not a bridge. Deleting it indeed leaves e_2 connecting u, v.

Skipping every edge to parent u would wrongly skip e_2,leave low[v]=2,and label e_1 a bridge. Correct edge-stack order pushes e_1 then e_2; low[v]≥disc[u] triggers one pop of e_2,e_1 as a two-edge block. Root u has one child and is not an articulation.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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